

p -ADIC SIMPSON CORRESPONDENCES VIA GERBES

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ABSTRACT. We introduce the Simpson gerbe attached to a smooth rigid space over a perfectoid field and discuss its basic properties. The p -adic Simpson correspondence of Faltings–Heuer can be reformulated in terms of twisted sheaves on this gerbe. This article is written for contribution to the proceeding of the workshop on “Complex and p -Adic Simpson Correspondence” in Maryland. All results are based on a joint work (in progress) with Bhargav Bhatt.

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1. INTRODUCTION

Classically, the Corlette–Simpson correspondence over the complex numbers, or nonabelian Hodge theory (see [Sim92]), states that on a compact Kähler manifold, there is an tensor equivalence between the category of $\mathbb{C}$ -local systems (equivalently, finite dimensional representations of the fundamental group) and that of semistable Higgs bundles with vanishing rational Chern classes. Faltings [Fal05], Deninger–Werner [DW05] first studied the p -adic analogue of this correspondence. These results were extended by [AGT16], [AG24] and many others. Recently, Heuer [Heu25] generalized Faltings’ result and established a p -adic Simpson correspondence for all smooth proper rigid spaces over algebraically closed p -adic fields.

To state his result, we let C be an algebraically closed non-archimedean field over $\mathbb{Q}_p$ with maximal ideal $\mathfrak{m}_C$, and X be a smooth proper rigid space over C . Consider Fontaine’s period ring $B_{\mathrm{dR}}^+(C)$, which is a complete discrete valuation ring. We pick a uniformizer t in it and write $B_2(C)$ for the quotient $B_{\mathrm{dR}}^+(C)/t^2$. Denote by X_v the v -site of X in the sense of [Sch22].

Theorem 1.1 (Faltings–Heuer). *Choose a flat lift $\tilde{X}$ of X to $B_2(C)^1$ and a continuous section to the p -adic logarithm map $\text{Exp} : C \rightarrow 1 + \mathfrak{m}_C$ (an exponential for C). Then there is a tensor equivalence, depending on the choices, between the categories of vector bundles on X_v and Higgs bundles² on $X_{\text{ét}}$.*

In the above statement, the v -site X_v can also be replaced by the pro-étale site $X_{\text{proét}}$, we use the former to be consistent with the lectures by Heuer and Reinecke. This is a big site of perfectoid spaces over X where coverings are generated by surjection of affinoid perfectoids. The presheaf $\mathcal{O}_{X_v}$ of rings sending a perfectoid $S \rightarrow X$ to $\mathcal{O}(S)$ is a sheaf, and a vector bundle on X_v is a locally free module for $\mathcal{O}_{X_v}$. Alternatively, via descent, a v -vector bundle can be viewed as a compatible collection of vector bundles on affinoid perfectoid spaces over X . We note that there is a fully faithful embedding of the category of C -valued finite dimensional continuous representations of the (pro-finite) étale fundamental group of X into v -vector bundles on X , using Scholze’s primitive comparison theorem [Sch12, Theorem 3.3]. This shows the analogy to the complex case and justifies naming the above theorem a “ p -adic Simpson correspondence”.

However, the p -adic correspondence is genuinely different from the complex one in two ways: On the one hand, the equivalence encompasses all Higgs bundles, without conditions on semi-stability or vanishing of the rational Chern classes. On the other hand, the equivalence depends on choices that are not present in the complex case. One of the main motivation of [BZ] is to understand these choices and formulate a more canonical p -adic Simpson correspondence that applies also to non-proper smooth rigid spaces. The question regarding which Higgs bundles correspond to representations of the fundamental group in the p -adic setting remains a mystery.

It turns out that for any smooth rigid space over a perfectoid field, there is an étale $\mathbf{G}_m$ -gerbe³ on its cotangent bundle, which we call the *Simpson gerbe*. Twisted sheaves (for this gerbe) on the cotangent bundle can be canonically related to sheaves on the v -site of the base space, and the choices in Theorem 1.1 are related to the choices of splittings of this gerbe on certain loci of the cotangent bundle.

This phenomenon is closely related to the nonabelian Hodge correspondence in characteristic p , established by Ogus–Vologodsky [OV07], see Petrov’s lecture. We briefly recall this, to compare with the p -adic case that we will explain later. Let k be a perfect field of characteristic p and X/k be a smooth scheme.

¹It always exists by spreading out, see [BMS18, Corollary 13.16, Proof of Theorem 13.19].

²We caution the reader that in the definition of Higgs bundles in the p -adic setting, the Kähler differential in the Higgs field has a Tate twist, see [Heu25, Definition 1.2].

³Recall that a $\mathbf{G}_m$ -gerbe is a sheaf of groupoids that has an action by the classifying stack $B\mathbf{G}_m$ and is étale locally (equivariantly) equivalent to $B\mathbf{G}_m$. In other words, it is a torsor under $B\mathbf{G}_m$. Note that $B\mathbf{G}_m$ parametrizes étale line bundles on test objects, which is clearly a stack for the étale topology.

Recall that in the complex case, via the Riemann–Hilbert correspondence, local systems on a Kähler manifold can also be viewed as vector bundles with flat connections. These are special cases of D -modules, namely quasi-coherent sheaves on X , which are also modules over the sheaf of crystalline differential operators $\mathcal{D}_X$. In characteristic p , the noncommutative sheaf of algebras $\mathcal{D}_X$ has a curious structure: it has a large center and is an Azumaya algebra over its center. More precisely, let us write $F : X \rightarrow X^{(1)}$ for the relative Frobenius of X/k ; let $\mathcal{T}_{X/k}$ be the tangent bundle of X , and $Z(F_*\mathcal{D}_X)$ be the center of the Frobenius pushforward of $\mathcal{D}_X$.

We can define the ($\mathcal{O}_{X^{(1)}}$ -linear) p -curvature map

$$\psi : \mathcal{T}_{X^{(1)}/k} \rightarrow F_*\mathcal{D}_X : 1 \otimes \partial \mapsto \partial^{\circ p} - \partial^{[p]},$$

where ∂ is a section of $\mathcal{T}_{X/k}$, $\circ$ is the composition in $\mathcal{D}_X$, and $\partial^{[p]}$ is the p -th iterate of the vector field ∂ , viewed as a derivation on X .⁴ The map ψ is clearly $\mathcal{O}_{X^{(1)}}$ -linear and it in fact lands in the center of $F_*\mathcal{D}_X$. We have the following result.

Theorem 1.2 ([BMR08]). *The p -curvature map ψ induces an isomorphism $\mathrm{Sym}^\bullet \mathcal{T}_{X^{(1)}/k} \simeq Z(F_*\mathcal{D}_X)$. Moreover, $F_*\mathcal{D}_X$ is an Azumaya algebra (over its center) of order p , and rank $p^{2 \dim X}$.*

In particular, we can consider the stack of trivializations of the Azumaya algebra $F_*\mathcal{D}_X$ and it gives rise to a $\mathbf{G}_m$ -gerbe $\mathcal{P}(X)$ (for the étale topology) on the cotangent bundle $T^*X^{(1)}$ of $X^{(1)}$. Moreover, we have a tensor equivalence

$$(1) \quad D\text{-Mod}(X) = \mathrm{Mod}_{\mathrm{QCoh}(T^*X^{(1)})}(F_*\mathcal{D}_X) \simeq \mathrm{QCoh}(T^*X^{(1)}, \mathcal{P}(X)).$$

Here the last term denotes the category of twisted (by $\mathcal{P}(X)$) sheaves on $T^*X^{(1)}$, i.e. those quasi-coherent sheaves on $\mathcal{P}(X)$ which are of degree 1 when pulled back under the action map $B\mathbf{G}_m \times \mathcal{P}(X) \rightarrow \mathcal{P}(X)$. It is equipped with a symmetric monoidal structure given by the convolution product. Note that if $\mathcal{P}(X)$ was a trivial gerbe, then the twisted sheaves would just be usual quasi-coherent sheaves on $T^*X^{(1)}$, or equivalently Higgs sheaves on $X^{(1)}$. This explains why Equation (1) itself is a twisted form of a Simpson correspondence.

Even though in reality $\mathcal{P}(X)$ is never trivial unless $\dim X = 0$, one can nevertheless trivialize $\mathcal{P}(X)$ on a thickening of the zero section in the cotangent bundle, after making suitable choices. This is the content of the nonabelian Hodge correspondence of Ogus–Vologodsky. To state this, we let $W_2(k)$ be the ring of truncated Witt vectors and define $\mathcal{D}_X^\gamma$ to be $\mathcal{D}_X \otimes_{\psi, F^*\mathcal{O}_{T^*X^{(1)}}} F^*\mathcal{O}_{T^*X^{(1)}, \#}$, where $T^*X^{(1), \#}$ is the divided power envelope of the zero section in $T^*X^{(1)}$.

⁴For a vector bundle with flat connection (E, ∇) on X , its p -curvature is the mapping $\mathcal{T}_{X/k} \rightarrow \underline{\mathrm{End}}_{\mathcal{O}_X}(E)$, $\partial \mapsto \psi(1 \otimes \partial)$. It measures the failure of $\nabla : \mathcal{T}_{X/k} \rightarrow \underline{\mathrm{End}}_k(E)$ to be a map of restricted Lie algebras. By a result of Cartier [Kat70, Theorem 5.1], it is also the obstruction of descending (E, ∇) along F .

Theorem 1.3 (Cf. [OV07, Theorem 2.8]). *Given a flat lift $\widetilde{X^{(1)}/W_2(k)}$ of X , one can trivialize $\mathcal{P}(X)$ on $T^*X^{(1),\sharp}$, the completion of the zero section in $T^*X^{(1),\sharp}$ with respect to the divided power filtration. This leads to a tensor equivalence*

$$\mathrm{D}\text{-}\mathrm{Mod}_\gamma^\bullet(X) \simeq \mathrm{QCoh}^\bullet(T^*X^{(1),\sharp}),$$

where $\mathrm{D}\text{-}\mathrm{Mod}_\gamma(X)$ is the category of $\mathcal{D}_X^\gamma$ -modules on X ; $\mathrm{QCoh}(T^*X^{(1),\sharp})$ is the category of PD-Higgs sheaves on $X^{(1)}$, and the superscript $(-)^{\bullet}$ means the full subcategory of locally PD-nilpotent objects.

Here the choice of a lift of $X^{(1)}$ provides an explicit splitting module for the restriction of the Azumaya algebra $F_*\mathcal{D}_X$ to the divided power completion of the zero section. Despite the reliance on this choice, the correspondence is natural in the lifts and is of local natural, in the sense that X needs not be proper.

It turns out that there are similar structures in the p -adic situation. Indeed, inspecting on Heuer's proof of the p -adic Simpson correspondence, a key ingredient is the multiplicative Hodge–Tate exact sequence

$$(2) \quad 0 \rightarrow \mathrm{Pic}(X_{\mathrm{\acute{e}t}}) \rightarrow \mathrm{Pic}(X_v) \xrightarrow{\mathrm{HTlog}} H^0(X, \Omega_{X/C}^1(-1)) \rightarrow 0,$$

which arises from the following computation of the higher direct images of $\mathbf{G}_m$ along the natural map $\nu : X_v \rightarrow X_{\mathrm{\acute{e}t}}$.

Proposition 1.4 ([Heu25, Proposition 2.9 (iii)]).

$$\nu_*\mathbf{G}_m \simeq \mathbf{G}_m; \quad R^i\nu_*\mathbf{G}_m \simeq \wedge^i(\Omega_{X/C}^1(-1)),$$

where $\Omega_{X/C}^1(-1)$ is the cotangent sheaf of X , with a Tate twist.

To simplify notation, let us write $\widetilde{\Omega}_X$ for the twisted cotangent bundle $\Omega_{X/C}^1(-1)$, and write $\widetilde{T}^*X$ for its total space. Then from above we obtain an exact triangle of complexes of sheaves of abelian groups on $X_{\mathrm{\acute{e}t}}$

$$(3) \quad \mathbf{G}_m \rightarrow R^{\leq 1}\nu_*\mathbf{G}_m \rightarrow \widetilde{\Omega}_X[-1].$$

In fact, the computation in Proposition 1.4 holds true for any smooth rigid space, not necessarily proper; though when X is proper, we have right exactness of Equation (2). In particular, we may apply Proposition 1.4 to the (twisted) cotangent bundle $\widetilde{T}^*X$ itself. Taking the global section of (3), we get a map from 1-forms on $\widetilde{T}^*X$ to the Brauer group of $\widetilde{T}^*X$. If we evaluate this map at the tautological 1-form, we get the class of a $\mathbf{G}_m$ -gerbe on the (twisted) cotangent bundle of X !

By considering Equation (3) on the big étale site of smooth rigid spaces over C , one can construct a natural $\mathbf{G}_m$ -gerbe $\mathcal{P}(X)$ giving rise to this class. This is what we call the *Simpson gerbe* for X . Using this gerbe, we can formulate a canonical p -adic Simpson correspondence in the spirit of Theorem 1.3, which is

of local natural: Let X/C be a smooth rigid space. We denote by $\mathrm{Vect}(X_v)$ the category of vector bundles on X_v , and by $\mathrm{Higgs}(X, \mathcal{P})$ the category of twisted Higgs bundles on $X_{\text{ét}}$, see Definition 3.1.

Theorem 1.5 ([BZ]). *There is a canonical tensor equivalence⁵*

$$\Phi_X : \mathrm{Higgs}(X, \mathcal{P}) \xrightarrow{\sim} \mathrm{Vect}(X_v).$$

Moreover, when X is proper, the choices of a lift of X to $B_2(C)$ and an exponential for C trivialize $\mathcal{P}(X)$ simultaneously on all spectral varieties and thus identifies $\mathrm{Higgs}(X, \mathcal{P})$ with the category of Higgs bundles on $X_{\text{ét}}$.

Here we use the notion of *spectral varieties* in the sense of [Heu25]. Namely, these are the supports of Higgs bundles on X , viewed as coherent sheaves on $\tilde{T}^*X$. Note that this is slightly different from the usual definition in the context of Hitchin fibration, see Footnote 1 of *loc. cit.*. Let us remark that if we are only given the $B_2(C)$ -lift, we can still trivialize $\mathcal{P}(X)$ over a small neighborhood of the zero section in $\tilde{T}^*X$, and get an equivalence between v -vector bundles and Higgs bundles on X with certain smallness conditions. This is similar to Theorem 1.3. In fact, the similarity to characteristic p is not a coincidence. In joint work in progress [BKAMZ], we show that in case X has a smooth formal model $\mathfrak{X}$ over $\mathrm{Spf} \mathcal{O}_C$, there is an integral $\mathbf{G}_m$ -gerbe $\mathcal{P}(\mathfrak{X})$ on $T^*\mathfrak{X}\{-1\}$ interpolating the gerbes $\mathcal{P}(X)$ and $\mathcal{P}(\mathfrak{X}_{\text{red}})$ in a suitable sense.

It is also worth mentioning is that, as a consequence of Theorem 1.5, we see that the category of v -vector bundles on X has a large center. At least, the algebra of global functions on $\tilde{T}^*X$ acts on $\mathrm{Vect}(X_v)$ through the above equivalence and its natural action on $\mathrm{Higgs}(X, \mathcal{P})$. This action actually upgrades to canonical maps $\nu^*\mathrm{Sym}^\bullet \tilde{\Omega}_X^\vee \rightarrow \underline{\mathrm{End}}(V)$ for each v -vector bundle V , or equivalently $\mathcal{O}_{X_v}$ -linear maps $\theta_V : V \rightarrow V \otimes \nu^*\tilde{\Omega}_X$. These are the canonical Higgs fields on v -vector bundles (or geometric Sen operators) constructed by Pan [Pan22] and Rodríguez-Camargo [Cam25a]. Whether they are zero or not, can be used to detect whether a v -vector bundle descends to the étale site, see the discussion in [Bha25, Section 12.5].

Under Cartier duality in the context of analytic stacks, the Simpson gerbe is dual to the analytic Hodge–Tate stacks of Anschütz, Le Bras, Scholze and Rodríguez-Camargo. This aspect is worked out in [Cam25b].

The rest of these notes will be devoted to explaining the construction of $\mathcal{P}(X)$ and Φ_X , their basic properties, and the main idea for proving Theorem 1.5. The result in [BZ] is verily proven for perfect complexes. For simplicity, we focus on the above non-derived statement.

⁵As we will explain later, the gerbe $\mathcal{P}(X)$ has a group structure and the symmetric monoidal structure on the left hand side is defined with respect to the convolution product.

2. SIMPSON GERBES

Let C over $\mathbb{Q}_p$ be a perfectoid field, with maximal ideal $\mathfrak{m}_C$, ring of integers $\mathcal{O}_C$ and residue field k . We introduce the notion of *smoothoid spaces* over C . This is the category of spaces for which we will construct the Simpson gerbes.

Definition 2.1 ([Heu24b, Definition 1.3]). An analytic adic space over C is called a smoothoid space if it locally admits a smooth morphism to some perfectoid space over C .

We will denote the category of smoothoid spaces over C by Sm_C , where the morphisms are morphisms of adic spaces. For each $X \in \mathrm{Sm}_C$, we define its *cotangent sheaf* (on $X_{\text{ét}}$) to be the conormal sheaf of the diagonal, i.e.

$$\Omega_{X/C}^1 := \Delta_{X/C}^*(\mathcal{I}/\mathcal{I}^2),$$

where $\Delta_{X/C} : X \hookrightarrow X \times X$ is the diagonal embedding and $\mathcal{I}$ is the ideal sheaf of the image. One can show that this is always finite locally free over $\mathcal{O}_X$, and is trivial if X is a perfectoid space over C . We can similarly define relative cotangent sheaf for maps between smoothoid spaces, and for composition of maps there are transitivity triangles between cotangent sheaves as usual.

We will omit the subscript $/C$ whenever we consider the cotangent sheaf relatively over C . Moreover, we will write $\tilde{\Omega}_X$ for the Tate-twisted cotangent sheaf $\Omega_X^1(-1)$, and write $\tilde{T}^*X$ for its total space.

Construction 2.2. Let X be a smoothoid space over C . For a smoothoid space S over X , we define $\mathcal{P}(X)(S)$ to be the groupoid whose objects are pairs $(\mathcal{L}, \omega)$, where $\mathcal{L}$ is a v-line bundle on S , ω is a global section of $f^*\tilde{\Omega}_X$, such that the image $\mathrm{HTlog}(\mathcal{L})$ of $\mathcal{L}$ under the Hodge–Tate logarithm map from Equation (2) agrees with the image of ω under $f^*\tilde{\Omega}_X \rightarrow \tilde{\Omega}_S$. Morphisms in $\mathcal{P}(X)(S)$ are automorphisms of $\mathcal{L}$.

We let $\mathcal{P}(X)$ be the presheaf of groupoids on the slice category $\mathrm{Sm}_{C/X}$ of smoothoid spaces over X , which sends $f : S \rightarrow X$ to $\mathcal{P}(X)(S)$. This presheaf is clearly a sheaf for the étale topology, and by construction it comes with a natural map

$$\psi : \mathcal{P}(X) \rightarrow \tilde{T}^*X, (\mathcal{L}, \omega) \mapsto \omega,$$

which is a quasi-torsor under $B\mathbf{G}_m$. It follows from exactness of Equation (3) that ψ is in fact surjective as étale sheaves, hence a $B\mathbf{G}_m$ -torsor, or equivalently a $\mathbf{G}_m$ -gerbe. We call $\mathcal{P}(X)$ the *Simpson gerbe* for X .

Remark 2.3. The above construction intends to describe the functor of points of the middle term (shifted by 1) in Equation (3). In fact, the gerbe $\mathcal{P}(X)$ should be thought of as the relative Picard functor $\underline{\mathrm{Pic}}(X_v/X_{\text{ét}})$, and the map ψ is a geometric form of the Hodge–Tate logarithm map HTlog .

Remark 2.4. Denote the projection $\tilde{T}^*X \rightarrow X$ by π . The class of $\mathcal{P}(X)$ in $H^2(\tilde{T}^*X, \mathbf{G}_m)$ can be described as follows: Consider the boundary map $\delta_X : \tilde{\Omega}_X \rightarrow \mathcal{O}_X^\times[2]$ of Equation (3). One can do the same construction for $\tilde{T}^*X$ itself, which is a smoothoid space. Restricting to the differentials coming from X gives a map of complexes

$$\pi^*\tilde{\Omega}_X \rightarrow \tilde{\Omega}_{\tilde{T}^*X} \xrightarrow{\delta_{\tilde{T}^*X}} \mathcal{O}_{\tilde{T}^*X}^\times[2].$$

In particular, we get a map

$$H^0(\tilde{T}^*X, \pi^*\tilde{\Omega}_X) \rightarrow H^2(\tilde{T}^*X, \mathbf{G}_m)$$

by taking global sections. On $\tilde{T}^*X$, the sheaf $\pi^*\tilde{\Omega}_X$ has a tautological section ω_{taut} . The class of the Simpson gerbe is the image of ω_{taut} under the above map.

Remark 2.5. When X is a non-smooth rigid space, the complex $R\nu_*\mathcal{O}_{X_v}^\times$ admits a similar description using the Deligne–Du Bois complex. Namely, it has an exhaustive filtration indexed by natural numbers, whose 0-th graded piece is $\mathcal{O}_X^\times$ and for $i \geq 1$, the i -th graded piece is $\underline{\Omega}_X^i(-i)[-i]$, where $\underline{\Omega}_X^*$ is the Deligne–Du Bois complex. Indeed, by the same computation as in [Heu25, Proposition 2.9 (iii)], this reduces to [Bha25, Theorem 7.4.2], cf. [Guo23]. However, note that each $\underline{\Omega}_X^i$ are themselves complexes (instead of being concentrated in a single degree), so in this case we do not have as nice a geometric interpretation of this result.

We list a few properties of the Simpson gerbe.

- (1) The gerbe $\mathcal{P}(X)$ is multiplicative, in the sense that the class of $\mathcal{P}(X)$ in $H_{\text{ét}}^2(\tilde{T}^*X, \mathbf{G}_m)$ comes from $\text{Ext}_X^2(\tilde{T}^*X, \mathbf{G}_m)$. This is because Equation (3) is an extension of complexes of sheaves of abelian groups on X . In particular, $\mathcal{P}(X)$ is canonically trivial over the zero section in $\tilde{T}^*X$, as

$$\begin{array}{ccc} B\mathbf{G}_m & \longrightarrow & 0_X \\ \downarrow & & \downarrow \\ \mathcal{P}(X) & \longrightarrow & \tilde{T}^*X \end{array}$$

is a pullback square.

- (2) The construction $X \mapsto \mathcal{P}(X)$ is functorial with respect to morphisms in Sm_C in the following sense: Given a map $f : Y \rightarrow X$ in Sm_C , there is a

commutative diagram

$$(4) \quad \begin{array}{ccccc} \mathcal{P}(Y) & \xleftarrow{b} & \mathcal{P}(Y \rightarrow X) & \xrightarrow{a} & \mathcal{P}(X) \\ \downarrow & & \downarrow & & \downarrow \\ \tilde{T}^*Y & \xleftarrow{\quad} & (\tilde{T}^*X)_Y & \xrightarrow{\quad} & \tilde{T}^*X \\ \downarrow & & \downarrow & & \downarrow \\ Y & \xlongequal{\quad} & Y & \xrightarrow{f} & X, \end{array}$$

where the top two squares and the bottom right one are all cartesian.⁶ Two important special cases of the functoriality are

- f is étale. Then $f^*\mathcal{P}(X) = \mathcal{P}(Y)$.
- Y is perfectoid. Then Ω_Y^1 , and hence $\mathcal{P}(Y)$, is trivial, so $\mathcal{P}(Y \rightarrow X) := f^*\mathcal{P}(X)$ is canonically trivialized.

- (3) If X is smooth over C of positive dimension, then the class of $\mathcal{P}(X)$ in $H_{\text{ét}}^2(\tilde{T}^*X, \mathbf{G}_m)$ is not torsion. In fact, since $\tilde{T}^*X$, and hence $\text{Ext}_X^2(\tilde{T}^*X, \mathbf{G}_m)$, has the structure of a $\mathbb{Q}_p$ -vector space, we only need to see the class of $\mathcal{P}(X)$ is nonzero. For this, by localizing on X , we can find an étale morphism $X \rightarrow \mathbb{T}^d$ (a *toric chart*), for a torus $\mathbb{T}^d := \text{Spa } C\langle T_i^\pm \rangle_{i=1, \dots, d}$, and by functoriality of the Simpson gerbe for étale maps mentioned above, we can reduce to the case $X = \mathbb{T}^d$. Take a smooth integral model $\mathfrak{X}$ of X over $\text{Spf } \mathcal{O}_C$, we construct in [BKAMZ] a $\mathbf{G}_m$ -gerbe $\mathcal{P}(\mathfrak{X})$ on $T^*\mathfrak{X}\{-1\}$, whose fiber over k is the gerbe $\mathcal{P}(\mathfrak{X}_k)$ coming from $\mathcal{D}_{\mathfrak{X}_k}$, cf. Theorem 1.2; and its adic generic fiber $\mathcal{P}(\mathfrak{X})_\eta$, when pushed out to $\mathbf{G}_m$, agrees with the restriction of $\mathcal{P}(X)$ to the subspace $(T^*\mathfrak{X}\{-1\})_\eta \subset \tilde{T}^*X$. In particular, since $\mathcal{P}(\mathfrak{X}_k)$ is nontrivial when $\dim X > 0$, $\mathcal{P}(\mathfrak{X})$ is a non-trivial gerbe. Since the natural map

$$H_{\text{ét}}^2(T^*\mathfrak{X}\{-1\}, \mathbf{G}_m) \rightarrow H_{\text{ét}}^2((T^*\mathfrak{X}\{-1\})_\eta, \mathbf{G}_m)$$

is injective, we obtain that the restriction of $\mathcal{P}(X)$ to this locus, and hence $\mathcal{P}(X)$ itself, is nonzero.

Remark 2.6. In particular, even if X comes from the analytification of an algebraic variety, the class of $\mathcal{P}(X)$ will not come from the algebraic Brauer group of the cotangent bundle of X , since the algebraic cotangent bundle of X is quasi-compact and all elements in its Brauer group are torsion. In contrast, $\tilde{T}^*X$ is not quasi-compact and this leads to the infinite order property of $\mathcal{P}(X)$. In fact, for any quasi-compact subspace $Z \subset \tilde{T}^*X$, the

⁶We use $\mathcal{P}(Y \rightarrow X)$ to denote the pullback of $\mathcal{P}(X)$ to Y (which is the same as the pullback of $\mathcal{P}(Y)$ to $\tilde{T}^*X_Y$). This is because the operation of pullback/pushforward on twisted Higgs sheaves that we will introduce later should remind the readers of similar operations on D-modules, where $\mathcal{D}_{Y \rightarrow X} := \mathcal{O}_Y \otimes_{f^{-1}\mathcal{O}_X} f^{-1}\mathcal{D}_X$ is a standard notation for the transfer bimodule.

restriction $\mathcal{P}(X)|_Z$ has finite order; and writing $T^*X \simeq X \times \mathbf{G}_a^d$ as the union $\cup_n T^*X_{\leq p^n}$ of balls around the zero section with increasing radii, the order of $\mathcal{P}(X)|_{\tilde{T}^*X_{\leq p^n}}$ grows as n grows. This raises the natural question⁷ whether fixing an integer n , the gerbe $\mathcal{P}(X)|_{\tilde{T}^*X_{\leq p^n}}$ is represented by an Azumaya algebra; if so, what that Azumaya algebra is explicitly.

- (4) When X is smooth and **proper** over C , one can trivialize $\mathcal{P}(X)$ simultaneously on all spectral varieties, i.e. the supports of Higgs bundles on X viewed as coherent sheaves on $\tilde{T}^*X$. The choices appearing in Theorem 1.1, namely a $B_2(C)$ -lift of X and an exponential for C , give rise to such a trivialization. A proof of this fact is given in Section 4, where we will also see that it allows one to deduce Theorem 1.1 from Theorem 1.5.

We learned the following example from Drinfeld and Heuer, which describes the Simpson gerbe explicitly for the analytification of tori. This is a fundamental example, since any smooth rigid space is locally étale over tori, and pullback of the Simpson gerbe along an étale map gives the Simpson gerbe. For simplicity, we just consider the 1-dimensional case.

Example 2.7. Let $X = \mathbf{G}_m$ be the multiplicative group. Consider the Kummer sequence

$$0 \rightarrow \mathbb{Z}_p(1) \rightarrow \widetilde{\mathbf{G}_m} \rightarrow \mathbf{G}_m \rightarrow 1,$$

where the middle term $\widetilde{\mathbf{G}_m} \sim \varprojlim_{x \mapsto x^p} \mathbf{G}_m$ ⁸ is the universal cover of $\mathbf{G}_m$. We denote its class in $\text{Ext}^1(\mathbf{G}_m, \mathbb{Z}_p(1))$ by α . On the other hand, we have the logarithm sequence on the additive group

$$(5) \quad 1 \rightarrow \mu_{p^\infty} \rightarrow 1 + \mathfrak{m}_C \mathbf{G}_a^+ \xrightarrow{\log} \mathbf{G}_a \rightarrow 0,$$

where the middle term is the rigid group representing the sheaf $1 + \mathfrak{m}_C \mathcal{O}^+$. It is an open subgroup of $\mathbf{G}_m$. Denote its class in $\text{Ext}^1(\mathbf{G}_a, \mu_{p^\infty})$ by β .

We twist the logarithm sequence by a (-1) -Tate twist to get $\beta(-1) \in \text{Ext}^1(\mathbf{G}_a(-1), \mu_{p^\infty}(-1))$. Then the class of the Simpson gerbe

$$[\mathcal{P}(X)] \in \text{Ext}^2(\tilde{T}^*X, \mathbf{G}_m) \simeq \text{Ext}^2(\mathbf{G}_m \times \mathbf{G}_a(-1), \mathbf{G}_m)$$

agrees with the image of the cup product $\alpha \cup \beta(-1)$, pushed out to $\mathbf{G}_m$.

Remark 2.8. This example gives another way of showing that for $X = \mathbf{G}_m$, $\mathcal{P}(X)$ is trivialized when pulled back to the universal cover $\widetilde{\mathbf{G}_m}$ (without using perfectoidness of the latter). Generally, if $X \rightarrow \mathbb{T}$ is a toric chart, and $\tilde{X} := X \times_{\mathbb{T}} \tilde{\mathbb{T}}$ is the pullback of the universal cover, then $\mathcal{P}(X)$ is trivialized when pulled back to $\tilde{X}$.

⁷This question was posed by Vadim Vologodsky.

⁸Here $\sim$ denotes the tilde-limit in the sense of [SW13, Definition 2.4.1].

3. THE TWISTED SIMPSON CORRESPONDENCE

Using the Simpson gerbe, we can state a local, canonical version of the p -adic Simpson correspondence. This requires us to replace the category of Higgs bundles on $X_{\text{ét}}$ by certain twisted sheaves on $\mathcal{P}(X)$. For simplicity, we let X/C be a smooth rigid space, instead of a more general smoothoid space.

Definition 3.1. A twisted Higgs bundle on $X_{\text{ét}}$, is a coherent sheaf⁹ on $\mathcal{P}(X)$ of degree 1, which, when pulled back along any map $S \rightarrow X$ from a perfectoid space, is a vector bundle over $\mathcal{O}_S$.

We denote by $\text{Higgs}(X, \mathcal{P})$ the category of twisted Higgs bundles.

Remark 3.2. Concretely, pick one (equivalently any) étale cover $\{U_i\}_{i \in I}$ of $\tilde{T}^*X$ and a trivialization $\alpha_i : \mathcal{P}(X)|_{U_i} \simeq B\mathbf{G}_m$ for each i . For each pair i and j , by comparing the trivializations on the intersection $U_{ij} := U_i \times_{\tilde{T}^*X} U_j$, we obtain a global section $\alpha_j \circ \alpha_i^{-1} \in B\mathbf{G}_m(U_{ij})$, which amounts to a line bundle $\mathcal{L}_{ij}$. Then an object $\mathcal{F} \in \text{Higgs}(X, \mathcal{P})$ can be identified with the data of a collection of coherent sheaves $\mathcal{F}_i$ on U_i , and isomorphisms $\varphi_{ij} : \mathcal{F}_i|_{U_{ij}} \xrightarrow{\sim} \mathcal{F}_j|_{U_{ij}} \otimes \mathcal{L}_{ij}$, such that the support¹⁰ of $\mathcal{F}$ is finite over X and each $\mathcal{F}_i$ is flat over X .

Since the gerbe $\mathcal{P}(X)$ is multiplicative, we have a convolution product, denoted by $*$, on the bounded derived category of coherent sheaves on $\mathcal{P}(X)$. The convolution product preserves $\text{Higgs}(X, \mathcal{P})$, making it a symmetric monoidal category.

Theorem 3.3 ([BZ]). *There is a canonical tensor equivalence*

$$\Phi_X : \text{Higgs}(X, \mathcal{P}) \xrightarrow{\sim} \text{Vect}(X_v).$$

Moreover, when X is proper, choosing a lift of X to $B_2(C)$ and an exponential for C , $\mathcal{P}(X)$ can be trivialized simultaneously on all spectral varieties and $\text{Higgs}(X, \mathcal{P})$ identifies with the category of Higgs bundles on $X_{\text{ét}}$.

Remark 3.4. In the case when X is proper, after making the suitable choices, if a Higgs bundle (E, θ) and v -vector bundle V correspond to each other under Φ_X , then one can also compare the Dolbeault cohomology of (E, θ) with the v -cohomology of V . This follows from the above theorem on the level of perfect complexes. We do not expand on this here.

To construct the functor Φ_X , following the intuition that the Simpson gerbe is the relative Picard functor of X_v over $X_{\text{ét}}$ (see Remark 2.3), morally we consider

⁹Coherent sheaves on a locally noetherian adic space is defined by Huber, see for example, [Joh24, Lecture 1]; coherent sheaves on $\mathcal{P}(X)$ makes sense via flat descent, see [Mat22, Theorem 7.8].

¹⁰Cf. [Lie07, Definition 2.2.6.2]

a Fourier–Mukai transform $p_{1,!}(p_2^* - \otimes \gamma^* \mathcal{O}(-1))$, for the diagram

$$\begin{array}{ccc} & \text{“} X_v \times_{X_{\text{ét}}} \mathcal{P}(X)_{\text{ét}} \text{”} & \xrightarrow{\gamma} BG_m \\ p_1 \swarrow & & \searrow p_2 \\ X_v & & \mathcal{P}(X)_{\text{ét}} \end{array}$$

where the map γ should come from the canonical splitting of $\mathcal{P}(X)$ on X_v , since the latter is locally perfectoid. While the fiber product in quote does not make literal sense (neither does $p_{1,!}$), inspired by this, one can make the following construction.

Construction 3.5. Let $\mathcal{F} \in \text{Higgs}(X, \mathcal{P})$ be a twisted Higgs bundle. We define $\Phi_X(\mathcal{F})$ to be the v -sheaf whose pullback to a perfectoid space $S \rightarrow X$ is the vector bundle $p_{1,S,*}(p_{2,S}^* \mathcal{F} \otimes \gamma_S^* \mathcal{O}(-1))$, with the maps as in the diagram below

$$\begin{array}{ccc} & S \times_X \mathcal{P}(X) & \xrightarrow{\gamma_S} BG_m \\ p_{1,S} \swarrow & & \searrow p_{2,S} \\ S & & \mathcal{P}(X). \end{array}$$

Here γ_S comes from the canonical splitting of $\mathcal{P}(X)$ over S . This indeed defines a vector bundle on X_v via v -descent of vector bundles, see [SW20, Lemma 17.1.8]. The assignment $\mathcal{F} \mapsto \Phi_X(\mathcal{F})$ is clearly functorial, so we obtain the desired functor Φ_X .

Remark 3.6. The fact that $\mathcal{P}(X)$ is canonically trivial when pulled back to X_v is an analogue of the fact that in characteristic p , the Azumaya algebra $F_* \mathcal{D}_X$ over $T^*X^{(1)}$ is canonically split when pulled back along the Frobenius $F : X \rightarrow X^{(1)}$, see [OV07, Theorem 2.1].¹¹

The functor Φ_X is symmetric monoidal for the convolution product on the source and usual tensor product on the target. This follows from the same argument as in [Lau96, Corollaire 1.3.3]. Namely, consider the multiplication on $\mathcal{P}(X)$ by

$$\mu : \mathcal{P}(X) \times_X \mathcal{P}(X) \rightarrow \mathcal{P}(X)$$

and the diagonal morphism

$$\Delta_{S/X} : S \rightarrow S \times_X S.$$

One can show that

$${}^L \Delta_{S/X}^* \circ \Phi \simeq \Phi \circ R\mu_*,$$

¹¹In fact, F factors through the natural map from X to its de Rham stack $X^{\text{dR}} \simeq (X^{(1)})^{\text{HT}}$, and $F_* \mathcal{D}_X$ already splits when pulled back to X^{dR} . Therefore the same construction as in Construction 3.5 will provide a Fourier–Mukai functor from twisted Higgs modules on $X^{(1)}$ to D-modules on X , giving the equivalence in Equation (1).

where on $\mathcal{P}(X) \times_X \mathcal{P}(X)$, Φ is defined similarly using the pairing obtained by post-composing $\gamma_S \times \gamma_S$ with the multiplication on $B\mathbf{G}_m$

$$S \times_X S \times_X \mathcal{P}(X) \times_X \mathcal{P}(X) \xrightarrow{\gamma_S \times \gamma_S} B\mathbf{G}_m \times B\mathbf{G}_m \xrightarrow{m} B\mathbf{G}_m.$$

Moreover, exterior tensors are preserved under Fourier–Mukai transform. Altogether, this shows the symmetric monoidality.

Example 3.7. For a Higgs bundle with zero Higgs field $(E, \theta = 0)$, since it is supported on the zero section in $\tilde{T}^*X$, where $\mathcal{P}(X)$ is canonically trivial, it can be viewed as a twisted Higgs bundle. We see that $\Phi_X(E, 0)$ is just the pullback of E to X_v .

The proof that Φ_X is an equivalence passes through “small correspondences”. Before sketching the proof idea, we first note that Φ_X is functorial with respect to pullback along étale morphisms, as well as pushforward along finite étale morphisms. We have the following lemma, which follows directly from the definition of Φ_X, Φ_Y .

Lemma 3.8. *Let $f : Y \rightarrow X$ be an étale morphism of smooth rigid spaces over C , then we have*

- (1) $\Phi_Y \circ f^* \simeq f^* \circ \Phi_X$.
- (2) If f is finite étale, then $f_* \circ \Phi_Y \simeq \Phi_X \circ f_*$.

In particular, to show Φ_X is an equivalence, we may étale localize on X by part (1) of the above lemma. Hence we may assume X is affinoid and admits a **finite étale** morphism to a smooth rigid space of good reduction, see proof of [Ach15, Theorem 6.1], cf. part (2), proof of [Bha25, Theorem 7.3.6]. Now using part (2) of Lemma 3.8, we may assume X has good reduction as well. We let $\mathfrak{X}$ be a smooth formal model of X over $\mathrm{Spf} \mathcal{O}_C$.

Bhatt–Lurie [BL22, Proposition 5.12] introduced the (relative) Hodge–Tate stack $\mathfrak{X}^{\mathrm{HT}}$ over $\mathfrak{X}$, which is a gerbe banded by the flat group scheme $T_{\mathfrak{X}/\mathcal{O}_C}^{\sharp}\{1\}$, i.e. the divided power envelope of zero in the tangent bundle of $\mathfrak{X}$, with a Breuil–Kisin twist by 1. Taking the relative Picard stack of $\mathfrak{X}^{\mathrm{HT}}$ over $\mathfrak{X}$, we obtain a $\mathbf{G}_m$ -gerbe over the completion of the (Breuil–Kisin twisted) cotangent bundle of $\mathfrak{X}$ along the zero section

$$\hat{\mathcal{P}}(\mathfrak{X}) := \underline{\mathrm{Pic}}(\mathfrak{X}^{\mathrm{HT}}/\mathfrak{X}) \rightarrow \widehat{T^*\mathfrak{X}}\{-1\}.$$

This is an integral analog of the Simpson gerbe, except that it does not live over the whole cotangent bundle of $\mathfrak{X}$. As mentioned in the introduction, in [BKAMZ], we show that one can indeed extend $\hat{\mathcal{P}}(\mathfrak{X})$ to a $\mathbf{G}_m$ -gerbe $\mathcal{P}(\mathfrak{X})$ over the whole $T^*\mathfrak{X}\{-1\}$, although $\hat{\mathcal{P}}(\mathfrak{X})$ suffices for our purpose here. We need to know the following two facts about $\mathcal{P}(\mathfrak{X})$:

- (1) It is compatible with the Simpson gerbe for X , in the sense that there is a natural $B\mathbf{G}_m$ -equivariant commutative diagram

$$(6) \quad \begin{array}{ccc} \hat{\mathcal{P}}(\mathfrak{X})_\eta & \longrightarrow & \mathcal{P}(X) \\ \downarrow & & \downarrow \\ (\hat{T}^*\mathfrak{X}\{-1\})_\eta & \longrightarrow & \tilde{T}^*X \end{array}$$

induced by taking the generic fiber of $\mathfrak{X}$.

- (2) The Fourier–Mukai transform for $\mathfrak{X}^{\text{HT}}$ versus $\hat{\mathcal{P}}(\mathfrak{X})$ is an equivalence, and is compatible with Φ_X .

The second point needs some explanation. First of all, by construction, there is a natural pairing

$$\beta : \mathfrak{X}^{\text{HT}} \times_{\mathfrak{X}} \hat{\mathcal{P}}(\mathfrak{X}) \rightarrow B\mathbf{G}_m$$

induced by the universal line bundle on the fiber product. Similar to Definition 3.1, we can define (nilpotent) twisted Higgs bundles on $\mathfrak{X}$ to be those degree 1 twisted quasi-coherent sheaves on $\hat{\mathcal{P}}(\mathfrak{X})$ that are Zariski locally on $\mathfrak{X}$ vector bundles over $\mathcal{O}_{\mathfrak{X}}$.¹² Let $\text{Higgs}(\mathfrak{X}, \hat{\mathcal{P}}(\mathfrak{X}))$ be the category of twisted Higgs bundles on $\mathfrak{X}$. Hence, we have a functor

$$\Phi_{\mathfrak{X}} = p_{1,*}(p_2^* \otimes \beta^* \mathcal{O}(-1)) : \text{Higgs}(\mathfrak{X}, \hat{\mathcal{P}}(\mathfrak{X})) \rightarrow \text{Vect}(\mathfrak{X}^{\text{HT}}),$$

where p_1, p_2 are the projection maps on $\mathfrak{X}^{\text{HT}} \times_{\mathfrak{X}} \hat{\mathcal{P}}(\mathfrak{X})$ to the first and second factors. By reducing to the case $\mathfrak{X}$ is affine, and hence $\hat{\mathcal{P}}(\mathfrak{X})$ is the split gerbe, one can show that this is an equivalence of categories that intertwines the convolution product on the left hand side with the tensor product on the right hand side.

On the other hand, there is a fully faithful functor (in fact on the level of perfect complexes, see [AHB25, Theorem 5.6], cf. [Bha25, Proposition 12.3.3] for a different proof)

$$\alpha_{\mathfrak{X}}^* : \text{Vect}(\mathfrak{X}^{\text{HT}})[\frac{1}{p}] \rightarrow \text{Vect}(X_v)$$

obtained by sending a vector bundle $\mathcal{F}$ on $\mathfrak{X}^{\text{HT}}$ to the v -vector bundle on X , whose restriction to each perfectoid space $\text{Spa}(R, R^+) \rightarrow X$ is the pullback of $\mathcal{F}$ along the induced map

$$\text{Spa}(R, R^+) \rightarrow \text{Spf } R^+ = (\text{Spf } R^+)^{\text{HT}} \rightarrow \mathfrak{X}^{\text{HT}}.$$

¹²Note that in contrary to the Simpson gerbe for X , $\mathfrak{X}^{\text{HT}}$ and hence $\hat{\mathcal{P}}(\mathfrak{X})$ is Zariski locally trivial on $\mathfrak{X}$, in fact, whenever $\mathfrak{X}$ is affine, see [BL22, Proposition 5.12]. Choosing a trivialization of $\hat{\mathcal{P}}(\mathfrak{X})$, twisted quasi-coherent sheaves on it identifies with coherent sheaves on $\tilde{T}^*\mathfrak{X}\{-1\}$, so this definition makes sense.

Then the compatibility mentioned in (2) is the statement that the diagram below is 2-commutative

$$\begin{array}{ccc}
\mathrm{Higgs}(\mathfrak{X}, \hat{\mathcal{P}}(\mathfrak{X}))[\frac{1}{p}] & \hookrightarrow & \mathrm{Higgs}(X, \mathcal{P}) \\
\downarrow \simeq \Phi_{\mathfrak{X}}[1/p] & & \downarrow \Phi_X \\
\mathrm{Vect}(\mathfrak{X}^{\mathrm{HT}})[\frac{1}{p}] & \xrightarrow{\alpha_{\mathfrak{X}}^*} & \mathrm{Vect}(X_v),
\end{array}$$

where the top arrow is induced by the compatibility of $\hat{\mathcal{P}}(\mathfrak{X})$ and $\mathcal{P}(X)$, namely Diagram (6).

The above shows that Φ_X identifies a full subcategory of $\mathrm{Higgs}(X, \mathcal{P})$ with a full subcategory of $\mathrm{Vect}(X_v)$. This is what we call the “small correspondence”. The terminology is easy to understand, since twisted Higgs bundles on X coming from $\mathrm{Higgs}(\mathfrak{X}, \hat{\mathcal{P}}(\mathfrak{X}))$ are supported on the generic fiber of $\widehat{T^*X}\{-1\}$, which is an open unit ball around the zero section in $\widehat{T^*X}$, a locus where the differentials are “small” in terms of p -adic valuations. Let us temporarily call the objects in the essential image of the two horizontal arrows “small objects”.

To conclude that Φ_X is an equivalence, it suffices to show that we can find an étale cover $f : U \rightarrow X$ such that the images of f^* on $\mathrm{Higgs}(X, \mathcal{P})$ (resp. $\mathrm{Vect}(X_v)$) lie in the subcategories spanned by small objects in $\mathrm{Higgs}(U, \mathcal{P})$ (resp. $\mathrm{Vect}(U_v)$); and so are the further pullback to $U \times_X U$. This would imply full faithfulness and essential surjectivity of Φ_X at the same time.

For twisted Higgs sheaves, this is easy. Namely, we may without loss of generality assume X admits an étale morphism to a torus $\mathbb{T} := \mathrm{Spa} C\langle T_i^{\pm} \rangle_{i=1, \dots, d}$, where $d = \dim X$. Then the multiplication by p map $[p]$ on $\mathbb{T}$ satisfies $[p]^* dT_i = pT_i^{p-1} dT_i$. That is, the differentials become p -adically closer to zero when pulled back along $[p]$. Hence by letting f be the pullback of $[p^n]$ to X , for a large enough integer n , f^* can make the twisted Higgs bundles on X small.

For v -vector bundles, this uses the argument of Faltings [Fal05, Lemma 1, Theorem 3]. Namely, given a v -vector bundle V on X of rank r , up to étale localizing on X , we may assume it has an $\mathcal{O}_{X_v}^+$ -lattice V^+ , see [Heu24a, Proposition 4.8]. Further localizing on $X_{\mathrm{ét}}$ if necessary, we may even assume that V^+/p^c is trivial as an $\mathcal{O}_X^+/p^c$ -bundle, for some large enough constant c . One may think of V^+ as a deformation of the trivial bundle $(\mathcal{O}_X^+/p^c)^{\oplus r}$. Since the latter lies in the image of $\alpha_{\mathfrak{X}}^*$, it suffices to compare the deformation theory of $(\mathcal{O}_{\mathfrak{X}^{\mathrm{HT}}}/p^c)^{\oplus r}$ and that of $(\mathcal{O}_X^+/p^c)^{\oplus r}$. In fact the cohomology groups controlling the two agree up to bounded p -power torsion, and $\alpha_{\mathfrak{X}}^*$ induces almost injective maps on these cohomology groups. Following Faltings, as long as $c > \frac{2}{p-1}$, one can argue inductively that for each positive integer n , there is an $\mathcal{O}_{\mathfrak{X}^{\mathrm{HT}}}/p^{c+n\epsilon}$ -bundle E_n , such that $V^+/p^{c+n\epsilon} \simeq \alpha_{\mathfrak{X}}^* E_n$ for some fixed constant $0 < \epsilon < (\frac{\epsilon}{2} - \frac{1}{p-1})/3$. Then

$E := \lim_n E_n$ maps to V^+ under $\alpha_{\mathfrak{X}}^*$. We leave the details to the interested reader, see also [AHB23, Corollary 3.27].

4. THE RELATION TO FALTINGS–HEUER’S CORRESPONDENCE

Let X/C be a smooth rigid space as before. As a last topic, we discuss how to recover Theorem 1.1 from Theorem 1.5. For this we need to relate the category of twisted Higgs bundles on X with that of genuine Higgs bundles on X , which would rely on splitting the gerbe $\mathcal{P}(X)$ compatibly on spectral varieties. This is possible when X is proper, after making the choices in the statement of Theorem 1.1. To explain this, let us first observe that when X is proper, for each individual spectral variety $Z \subset \tilde{T}^*X$, the gerbe $\mathcal{P}(X)|_Z$ is trivial. We first introduce two lemmas, which themselves do not use properness.

Let $\mu_{p^\infty} \subset \mathbf{G}_m$ be the p^∞ -torsion subgroup. Since the characteristic of C is zero, this is also the étale sheaf $\mathbb{Q}_p/\mathbb{Z}_p(1)$ on the big étale site of C . Denote the natural projection from the v -site of X to the étale site by $\nu : X_v \rightarrow X_{\text{ét}}$.

Lemma 4.1. *For any open subgroup U of $\mathbf{G}_m$, $\mathcal{P}(X)$ has a canonical reduction of structure group to U .*

Proof. It suffices to consider subgroups of the form $U = 1 + p^n \mathcal{O}_X^+$ for all integer n ’s, where this follows from the computation of $R^{\leq 1} \nu_*(1 + p^n \mathcal{O}_{X_v}^+)$, see [Heu24a, Theorem 1.7]. $\square$

Denote the projection from $\tilde{T}^*X$ to X by π .

Lemma 4.2. *A flat lift $\tilde{X}$ of X to $B_2(C)$ induces a reduction of structure group of $\mathcal{P}(X)$ to μ_{p^∞} .*

Proof. Applying Lemma 4.1 to $U = 1 + \mathfrak{m}_C \mathcal{O}^+$ on $\tilde{T}^*X$, we obtain a reduction of $\mathcal{P}(X)$ to a $1 + \mathfrak{m}_C \mathbf{G}_a^+$ -gerbe, whose classifying map fits in a commutative diagram, cf. Remark 2.4

$$\begin{array}{ccc} \tilde{\Omega}_{\tilde{T}^*X} & \longrightarrow & 1 + \mathfrak{m}_C \mathcal{O}_{\tilde{T}^*X}^+[2] \\ \uparrow & \nearrow \mathcal{P}(X) & \downarrow \log \\ \pi^* \tilde{\Omega}_X & \longrightarrow & \pi^* \mathcal{O}_X = \mathcal{O}_{\tilde{T}^*X}[2] \end{array}$$

Now the lift $\tilde{X}$ provides a null homotopy of the bottom horizontal arrow, since it defines a Faltings extension, see [Heu25, Remark 2.17], which by adjunction corresponds to a section to $R^{\leq 1} \nu_* \mathcal{O}_{X_v} \rightarrow \tilde{\Omega}_X$ ¹³. Thus we obtain a lift of the

¹³In fact, the datum of a lifting and that of a splitting of the sequence

$$\mathcal{O}_X \rightarrow R^{\leq 1} \nu_* \mathcal{O}_{X_v} \rightarrow \tilde{\Omega}_X[-1]$$

are equivalent, since both form torsors under $H^1(X, T_X(1))$.

arrow labeled by $\mathcal{P}(X)$ to the fiber of $\log$, which is $\mu_{p^\infty}[2]$. This gives the reduction of structure group as wanted. $\square$

Using Lemma 4.2, we can rephrase the result in [Heu25, Section 2.6] as below.

Proposition 4.3. *If X is a smooth and proper rigid space over C and if $Z \subset \tilde{T}^*X$ is the support of a twisted Higgs bundle on X , then the pullback of $\mathcal{P}(X)$ to Z is (non-canonically) split.*

Proof. Denote the structure map from Z to X by f . Choose a lift $\tilde{X}$ of X to $B_2(C)$ to get a reduction of structure group of $\mathcal{P}(X)$ to a μ_{p^∞} -gerbe $\mathcal{P}'$. By taking sections of its classifying map $\tilde{T}^*X \rightarrow \mu_{p^\infty}[2]$ over Z , we obtain a map of sheaves on $\mathrm{Sm}_{C,\mathrm{\acute{e}t}}$

$$\partial : \underline{\mathcal{H}}^0(Z, f^*\tilde{\Omega}_X) \rightarrow \underline{\mathcal{H}}^2(Z, \mu_{p^\infty}),$$

where the two sides are the sheafification of the presheaves $S \mapsto H^0(Z \times S, pr_1^* f^* \tilde{\Omega}_X)$, and $S \mapsto H^2(Z \times S, \mu_{p^\infty})$. By the projection formula and properness of Z , the former is represented by the rigid group $H^0(Z, f^* \tilde{\Omega}_X) \otimes \mathbf{G}_a$, and the later is the constant sheaf $\underline{H}^2(Z, \mu_{p^\infty})$. But this means the map ∂ has to be zero, since a map from a connected space to a discrete space is constant. Evaluating ∂ on $\mathrm{Spa} C$, we see that the class of $\mathcal{P}'|_Z$ must be zero. $\square$

Remark 4.4. Here it is crucial that $H^0(Z, f^* \tilde{\Omega}_X)$ is finite dimensional over C , which uses properness, otherwise completion will be involved in the identification

$$H^0(Z \times S, pr_1^* f^* \tilde{\Omega}_X) \simeq H^0(Z, f^* \tilde{\Omega}_X) \hat{\otimes} \mathcal{O}(S).$$

This observation implies that fixing a spectral variety Z , we can assign a v -vector bundle to a Higgs bundle supported in Z , by first choosing a splitting of $\mathcal{P}(X)|_Z$, and hence viewing the Higgs bundle as a twisted Higgs bundle, and then appealing to Theorem 1.5. In fact, a much stronger result is true. Namely, splittings of $\mathcal{P}(X)|_Z$ can be chosen for all Z 's simultaneously and compatibly with intersections between them. Heuer's original proof of this fact relies on Proposition 4.3, and rigidifications of the splittings of $\mathcal{P}(X)|_Z$ on individual Z 's. The proof below does not rely on these considerations, and implies Proposition 4.3 as a consequence.

Proposition 4.5. *The choices of a lift $\tilde{X}$ of X to $B_2(C)$ and an exponential $\mathrm{Exp} : C \rightarrow 1 + \mathfrak{m}_C$ for C lead to compatible splittings of the restrictions of $\mathcal{P}(X)$ to all spectral varieties.*

Proof sketch. Let $Z \subset \tilde{T}^*X$ be a spectral variety and denote its structure map to X by f . We need to show that the choices induce a natural (in Z) splitting of the exact sequence

$$(7) \quad R^{\leq 1} \Gamma(X_{\mathrm{\acute{e}t}}, f_* \mathcal{O}_Z^\times) \rightarrow R^{\leq 1} \Gamma(X_v, (\nu^* f_* \mathcal{O}_Z)^\times) \rightarrow H^0(X_{\mathrm{\acute{e}t}}, f_* \mathcal{O}_Z \otimes \tilde{\Omega}_X)[-1].$$

We explain the idea of the proof when Z is smooth¹⁴. In this case, this sequence is the pullback of the multiplicative Hodge–Tate sequence for Z

$$R^{\leq 1}\Gamma(Z_{\text{ét}}, \mathcal{O}_Z^\times) \rightarrow R^{\leq 1}\Gamma(Z_v, \mathcal{O}_{Z_v}^\times) \rightarrow H^0(Z_{\text{ét}}, \tilde{\Omega}_Z)[-1]$$

along the natural injective map

$$\gamma : H^0(X_{\text{ét}}, f_*\mathcal{O}_Z \otimes \tilde{\Omega}_X) \simeq H^0(Z_{\text{ét}}, f^*\tilde{\Omega}_X) \rightarrow H^0(Z_{\text{ét}}, \tilde{\Omega}_Z),$$

so it suffices to provide a map $H^0(X_{\text{ét}}, f_*\mathcal{O}_Z \otimes \tilde{\Omega}_X)[-1] \rightarrow R^{\leq 1}\Gamma(Z_v, \mathcal{O}_{Z_v}^\times)$ lifting γ .

Denote by $\underline{C}$ the v -sheaf attached to the topological group C , which sends a perfectoid space S to continuous maps from the underlying topological space of S to C , similarly for $1 + \mathfrak{m}_C$ in place of C . Then the desired lifting will come from a composition of maps

$$\begin{aligned} H^0(X_{\text{ét}}, f_*\mathcal{O}_Z \otimes \tilde{\Omega}_X)[-1] &\xrightarrow{a} R^{\leq 1}\Gamma(Z_v, \mathcal{O}_{Z_v}) \xrightarrow{\sim} R^{\leq 1}\Gamma(Z_v, \underline{C}) \\ &\xrightarrow{\text{Exp}} R^{\leq 1}\Gamma(Z_v, 1 + \mathfrak{m}_C) \rightarrow R^{\leq 1}\Gamma(Z_v, \mathcal{O}_{Z_v}^\times), \end{aligned}$$

where the second map is given by the primitive comparison theorem [Bha25, Corollary 7.3.7] (which uses properness though not smoothness of Z); the third map is induced by the chosen exponential for C , and the third map is induced by the natural injection $1 + \mathfrak{m}_C \rightarrow \mathcal{O}_{Z_v}^\times$.

To construct the map a , note that the choice of $\tilde{X}$ splits the complex $R\nu_*\mathcal{O}_{X_v}$ and hence also its tensor with $f_*\mathcal{O}_Z$. This in particular leads to a splitting of the following sequence

$$R^{\leq 1}\Gamma(X_{\text{ét}}, f_*\mathcal{O}_Z) \rightarrow R^{\leq 1}\Gamma(X_v, \nu^*f_*\mathcal{O}_Z) \rightarrow H^0(X_{\text{ét}}, f_*\mathcal{O}_Z \otimes \tilde{\Omega}_X)[-1].$$

We let a be the composition of this splitting with the natural base-change map

$$R^{\leq 1}\Gamma(X_v, \nu^*f_*\mathcal{O}_Z) \rightarrow R^{\leq 1}\Gamma(Z_v, \mathcal{O}_{Z_v}).$$

When Z is non-smooth, the proof needs to be modified, since on the one hand, the multiplicative Hodge–Tate sequence for Z involves the Deligne–Du Bois complex, see Remark 2.5. While on the other hand, the sequence labeled (7) is no longer a pullback of the multiplicative Hodge–Tate sequence for Z (unless Z has Du Bois singularities, in which case $\mathcal{O}_Z \simeq \underline{\mathcal{O}}_Z$). We refer to [Bha25, Theorem 12.5.7] for a detailed treatment in this case. $\square$

As a corollary, we recover the result of Faltings and Heuer.

Corollary 4.6 (Theorem 1.1). *The choices $\tilde{X}$ and Exp lead to a tensor equivalence between v -vector bundles on X and Higgs bundles.*

¹⁴This proof in this case is suggested by Sasha Petrov.

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